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Speaker Notes — Progress Presentation

Companion to thesis/slides.pdf (18 slides). English lines are what you say out loud; [TR] notes are delivery guidance for you only.

Delivery plan

- **Target 16–18 min** + questions. Spend **2 min on slide 4 (cascade architecture)**, **2 min on slide 10 (normalisation)** and **2 min on slide 11 (why replay still underperforms)** — the three things they asked for.
- **[TR]** Üç ana çıktı: (1) cascade mimarisi — hocaların istediği, (2) normalizasyon replay’i kurtardı, (3) replay’in neden hâlâ tam çözülmediğini somut kanıtla anlatıyoruz. Bu üçünde yavaşla.
- **One sentence to remember:** *“We built the interpretable cascade you asked for, the normalisation idea recovered replay ranking, and we can now show concretely why the last bit of replay is a physical limit, not a bug.”*

Slide 1 — Title

“Thanks. Three things today: the new cascade architecture, a normalisation fix that recovers replay, and concrete evidence for why replay still isn’t fully solved.”

Slide 2 — The problem

“A water network is a graph — junctions report pressure, pipes report flow. Sensing is sparse, half the readings missing, and noisy. An attacker falsifies a subset of sensors. Five attack families. The detector reconstructs the truth and flags which sensors lie.”

[TR] “Stealthy” ve “Replay”i vurgula.

Slide 3 — Pipeline

“Network, simulate, inject attacks, detect, evaluate. Only the first stage is domain-specific — the rest is shared code, which matters for Part 2.”

Slide 4 — The architecture: cascade routing (2 min)

[TR] Hocaların istediği mimari. Diyagramı soldan sağa elinle takip et.

“This is the architecture. Instead of blending experts opaquely, the router ranks them and we run one at a time. Snapshots go to the router — a fast classifier — which ranks the experts. We run the top one, then a feedback check asks: did it do a good job? On negative feedback we go back and try the next expert — the orange loop — until one is accepted. Every prediction carries which expert ran and why it was kept or rejected. It’s interpretable.”

Slide 5 — The feedback mechanism

“The feedback is label-free, because at inference we have no ground truth. Two signals: do the sensors the expert calls ‘clean’ agree with their readings, and does the reconstruction conserve mass? We standardise the two terms so neither dominates, restrict re-routing to the router’s top-2 for stability, and trust a confident router unless the feedback clearly disagrees.”

[TR] Formülü işaret et, okuma. “consistency + physics, etiketsiz” de.

Slide 6 — The experts

“The experts the cascade routes among: six, each a GraphSAGE-plus-GRU model over the 6-step window. Two training choices matter — direct expert supervision, so every expert learns its own class regardless of routing and never starves; and a physics loss that forbids hydraulically impossible reconstructions. These same experts become the substrate for power and traffic in Part 2, unchanged.”

[TR] Hızlı geç, ~45 saniye.

Slide 7 — Cascade vs the alternatives

“Over 5 seeds the cascade reaches 0.772, matching the oracle ceiling — the best any hard routing could do — and within a point of a full soft blend. The experts are similar here, so the interpretable cascade costs almost no accuracy, and it’s stable.”

[TR] “matches the ceiling, stable, costs almost no accuracy” — üçünü vurgula.

Slide 8 — Detection results

“Detection over 5 seeds with a calibrated threshold. Four of five attacks caught reliably. Replay near zero — the next three slides are why, how we recovered most of it, and why it’s still not solved.”

Slide 9 — Router diagnostic

“Is the router mis-routing? On Modena it’s healthy, 96%. On Net3, 77% of stealthy windows go to the replay expert — both look smooth, 97 nodes too few to separate them. The cascade feedback targets exactly this.”

Slide 10 — Recovering replay: normalisation (2 min)

[TR] Hocaların “normalize dene” önerisi. Yavaşla.

“Your hypothesis: water values sit too close together. We tested it. Normalising each sensor by its own temporal scale amplifies the tiny per-node variation a replay lacks. Middle panel: replay ranking jumps from below chance — 0.46 — to AUROC 0.79. Overall F1 holds at 0.833. The normalisation idea worked.”

Slide 11 — Why replay still underperforms (2 min)

[TR] Bu yeni slayt. Somut kanıt. Yavaşla, iki paneli tek tek göster.

“But replay’s F1 is still low, and here’s concretely why. Left: the anomaly-score distributions. Replayed sensors — purple — do score higher than genuine ones — green — so the ranking works, AUROC 0.75. But the two distributions overlap heavily. Right: because they overlap, no threshold gives high precision and recall together — the best F1 is 0.41.

The reason is physical. A replay is a verbatim copy of a past reading, so the only thing distinguishing it from a genuine value is the fresh observation noise it lacks. That residual signal is real but weak — this is an information limit, not a training bug.”

[TR] “overlap □ no threshold separates them □ physical limit” zincirini net kur. Bu, “neden hâlâ çözülmedi” sorusunun kanıtlı cevabı.

Slide 12 — Part 2 divider

“Does any of this survive outside water?”

Slide 13 — One detector, three physics

“Same detector, unchanged, on a power grid — IEEE 118-bus — and a road-traffic network. Voltage and speed instead of pressure. Only a data adapter changed.”

Slide 14 — Cross-domain F1

“It holds — F1 0.82 to 0.89 across three unrelated physical systems, no tuning.”

Slide 15 — The normalisation fix generalises

“Two messages in one figure. The grey bars — global normalisation — already track signal speed: replay is hardest in slow water (0.46) and trivial in fast traffic (0.97). The purple bars show per-node normalisation lifts it most where it’s hardest — water 0.46 to 0.79, power 0.66 to 0.90 — and isn’t needed where it’s already easy. A principled cross-domain fix, not a water-specific hack.”

[TR] Grey vs purple ayrımını net anlat — hem signal-speed hem fix tek figürde.

Slide 16 — Difficulty inversion

“And the hard case flips: water and power find stealthy easy, replay hard; traffic is the mirror image. The domain, not the model, decides.”

Slide 17 — Summary

“Part 1: the interpretable cascade matches the ceiling; the router failure diagnosed; per-node normalisation recovered replay ranking; and we showed the residual replay gap is a physical limit. Part 2: the detector transfers unchanged, the normalisation fix generalises, and replay difficulty tracks signal speed. Next steps: push the cascade where experts differ more, extend the threat model, pick a venue. I’d value your input on priority.”

Slide 18 — Thank you

“Thank you — questions?”

Anticipated questions & answers

On the cascade

Q: Why not just use the soft mixture — isn’t it slightly higher?

“It is, by about a point, but it’s an opaque blend of all six experts. The cascade gives essentially the same accuracy — it matches the oracle ceiling — while telling you which expert ran and why. For a safety-critical detector, that interpretability is worth more than a point of F1.”

Q: How stable is the cascade?

“0.772 plus or minus 0.017 over 5 seeds. Stability comes from restricting re-routing to the router’s top-2 experts, so the feedback can’t land on a wildly wrong one.”

Q: What is the feedback signal exactly?

“Two label-free terms: reconstruction agreement on the sensors the expert calls clean, and mass-conservation of the reconstructed flow. Z-scored across experts so neither dominates.”

On replay (the key evidence)

Q: Why is replay F1 still low if AUROC is 0.79?

“That’s exactly slide 11. AUROC measures ranking — replayed sensors do score higher. But the two score distributions overlap, so no single threshold separates them cleanly; the best F1 is 0.41. It’s the classic high-AUROC-low-F1 situation.”

Q: Could a better feature fix it?

“We tested the most direct one — an exact-lag-match detector, since a replay is a verbatim copy. On its own it only reached AUROC 0.62, because with 50% missing data and replays that reach outside the 6-step window, the exact match often isn’t visible. The model already captures most of the available signal.”

Q: So is replay unsolvable?

“Near its information limit on slow-signal domains, yes — the only cue is the missing observation noise of a copy. On fast-signal domains like traffic it’s already easy (AUROC 0.97). The domain’s signal dynamics decide, not the model.”

On normalisation / cross-domain

Q: Does per-node normalisation add information?

“No — it rescales each sensor by its own variance so the small signal that is there isn’t swamped. AUROC going 0.46 to 0.79 shows the information was present; the model just couldn’t exploit it under a global scale.”

Q: Why did power’s overall F1 drop under per-node?

“The same replay-recall trade-off as water: recovering replay costs a little precision elsewhere. Water and traffic actually gained overall F1; power traded about three points for a big replay-ranking gain.”

Q: Is the traffic data real?

“Synthetic road-sensor graph; the power grid is a standard IEEE-118 benchmark. A real traffic benchmark like METR-LA is the natural next step — only the adapter changes.”

On the thesis

Q: Contribution / venue?

“An interpretable cascade detector for physics-constrained sensor graphs; a diagnosis of the routing and replay limits with concrete evidence; a normalisation fix that generalises across domains. Fits CPS-security venues — I’d like to choose one with you.”

If you get a stuck question

1. *“Fair point — we haven’t tested that yet; good candidate for the next run.”*
2. *“I’d want to check the numbers before answering properly — can I come back?”*
3. *“I’m not sure — what would you expect to see?”*

Never invent a number. Unsure □ say “about 0.79”.